

RetailHub — Business Analysis Report

Week 1 Data Analytics Assignment — Data Cleaning, EDA & Business Insights

1. Executive Summary

This report summarizes the cleaning and analysis of RetailHub's customer, product, order, payment, and category datasets. The raw data contained substantial missing values (~7.95% of all cells), 38 duplicate rows across the five files, inconsistent text formatting, and invalid negative/zero/sentinel values in numeric columns. After cleaning and merging, a final sales table of 133 valid orders was produced and loaded into a SQLite database for querying.

2. Data Quality Findings

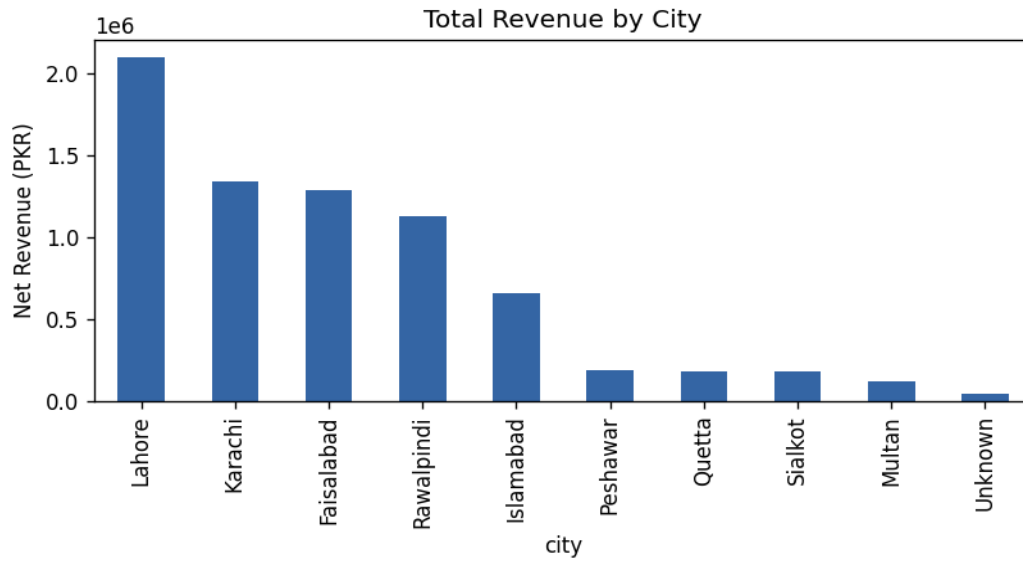
Dataset	Raw Rows	Duplicates Removed	Missing Cells (before)	Missing Cells (after)
Customers	139	9	64	0
Orders	146	7	95	0
Products	141	7	113	0
Payments	150	6	67	0
Categories	163	9	33	0

Additional invalid records handled: 6 orders had impossible customer/product IDs (sentinel values such as 9999/8888) and were dropped; 1 payment record had no order reference and was dropped; negative ages, prices, stock, quantities, and amounts were treated as invalid and either recomputed or imputed with the median/appropriate category value.

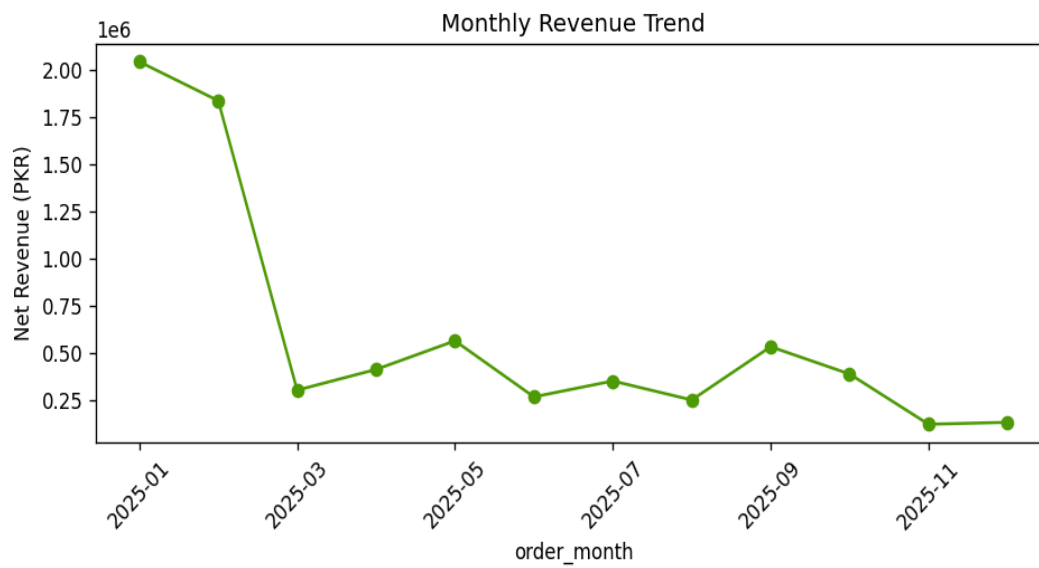
3. Key Business Metrics

Metric	Value
Total Revenue	PKR 7,204,923.99
Average Order Value	PKR 54,172.36
Total Customers	79
Total Orders	133
Top City by Revenue	Lahore (PKR 2,098,376.55)
Top Category by Revenue	Category 35 (PKR 1,186,189.61)
Best-Selling Product (units)	Product 20 (26 units)
Highest Revenue Month	January 2025 (PKR 2,042,098.88)
Most Common Payment Method	Cash (36 of 134 known-method payments)
Pending Payments (all records)	37
Top Customer by Spend	Customer 35 (PKR 476,333.15)

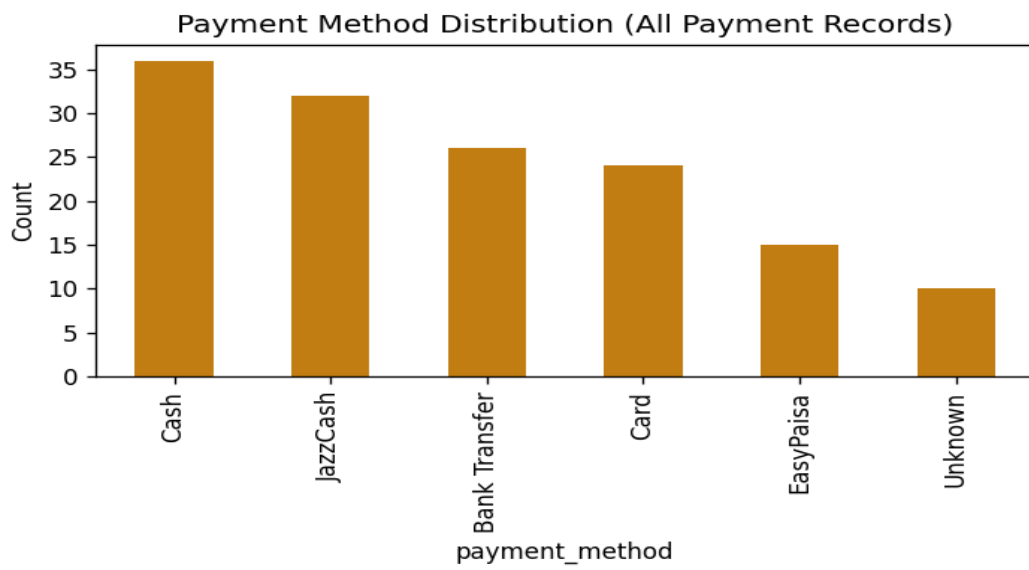
4. Visualizations



Total Revenue by City — Lahore leads, followed by Karachi and Faisalabad.



Monthly Revenue Trend — January 2025 was the strongest month.



Payment Method Distribution across all 134 known-method payment records — Cash is the most used method.

5. Answers to Business Questions

1. Highest revenue city: Lahore.
2. Best-performing category: Category 35.
3. Best-selling product by units: Product 20 (26 units).
4. Highest sales month: January 2025.
5. Most commonly used payment method: Cash (36 uses), ahead of JazzCash (32), Bank Transfer (26), Card (24), and EasyPaisha (15) — measured across all 134 known-method payment records, not just the subset tied to orders that survived cleaning.
6. Missing values in raw data: approximately 7.95% of all cells.
7. Duplicate records removed: 38 across all five datasets.
8. Highest spending customer: Customer 35 (PKR 476,333.15).
9. Additional information that would improve future analysis: return/refund records, customer acquisition channel, product cost price (for margin analysis), delivery time, and customer satisfaction ratings.

6. Recommendations

Focus marketing spend on Lahore and Karachi where revenue concentration is highest. Investigate the 45 orders with no linked payment record — this represents a data-integrity gap in the payments pipeline. Standardize data capture (city names, payment method labels, gender values) at the point of entry to reduce future cleaning overhead.